

Sheth L.U.J & Sir M.V College of Science

Subject :- Artificial Intelligence

Practical no 5

Aim :- Adaboost Ensemble Learning

- **Implement the Adaboost algorithm to create an ensemble of weak classifiers.**
- **Train the ensemble model on a given dataset and evaluate its performance.**
- **Compare the results with individual weak classifiers.**

Description :-

Step	Code / Function	Description
1	<code>pd.read_csv()</code>	Loads the Kaggle Phishing Website Detection dataset.
2	<code>dropna()</code>	Removes missing values from the dataset.
3	<code>drop_duplicates()</code>	Removes duplicate records.
4	X and y	Separates website features and the CLASS_LABEL target.
5	<code>pd.get_dummies()</code>	Converts categorical features into numerical values.
6	<code>train_test_split()</code>	Divides the dataset into 80% training and 20% testing data.

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7	DecisionTreeClassifier(max_depth=1)	Creates a single weak classifier called a Decision Stump.
8	weak_classifier.fit()	Trains the individual weak classifier.
9	AdaBoostClassifier()	Creates an ensemble of multiple weak classifiers.
10	n_estimators=50	Uses 50 weak decision trees in the AdaBoost ensemble.
11	ada_model.fit()	Trains the AdaBoost model by giving more importance to incorrectly classified samples.
12	predict()	Predicts whether websites are legitimate or phishing.
13	accuracy_score()	Calculates the accuracy of both models.
14	classification_report()	Displays precision, recall, and F1-score for AdaBoost.
15	confusion_matrix()	Shows correct and incorrect classifications.

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16	<code>plt.imshow()</code>	Visualizes the AdaBoost confusion matrix.
17	<code>plt.bar()</code>	Compares the accuracy of the weak classifier and AdaBoost.
18	<code>improvement</code>	Calculates how much AdaBoost improves over the individual weak classifier.

Code :-

```
T092 Practical no 6.py - C:/Users/Vinod Mali/Desktop/AI/T092 Practical no 6.py (3.14.6)
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix

df = pd.read_csv("dataset6.csv")

print("Dataset loaded successfully!")

print("\nDataset Shape:")
print(df.shape)

print("\nFirst 5 Rows:")
print(df.head())

df = df.dropna(axis=1, how="all")

df = df.drop_duplicates()

df = df.dropna()

print("\nShape after cleaning:")
print(df.shape)
```

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```
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File Edit Format Run Options Window Help
target_column = None

for column in df.columns:
    if column.strip().upper() == "CLASS_LABEL":
        target_column = column
        break

if target_column is None:
    print("\nAvailable columns:")
    print(df.columns.tolist())

    raise ValueError(
        "CLASS_LABEL column was not found."
    )

print("\nTarget Column:")
print(target_column)

X = df.drop(
    target_column,
    axis=1
)

y = df[target_column]

X = pd.get_dummies(
    X,
    drop_first=True,

```

```
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    dtype=int
)

y = pd.to_numeric(
    y,
    errors="coerce"
)

valid_rows = y.notna()
X = X.loc[valid_rows]
y = y.loc[valid_rows]
y = y.astype(int)

print("\nNumber of Features:")
print(X.shape[1])

print("\nTarget Classes:")
print(y.value_counts())

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,

```

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    stratify=y
)

print("\nTraining Samples:")
print(len(X_train))

print("\nTesting Samples:")
print(len(X_test))

weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=42
)

weak_classifier.fit(
    X_train,
    y_train
)

weak_prediction = weak_classifier.predict(
    X_test
)

weak_accuracy = accuracy_score(
    y_test,
    weak_prediction
)

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```

```
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)

print("\n=====")
print("INDIVIDUAL WEAK CLASSIFIER")
print("=====")

print(
    "Accuracy:",
    round(weak_accuracy * 100, 2),
    "% "
)

ada_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(
        max_depth=1,
        random_state=42
    ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)

print("\nTraining AdaBoost...")

ada_model.fit(
    X_train,
    y_train
)

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```

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print("AdaBoost trained successfully!")

ada_prediction = ada_model.predict(
    X_test
)

ada_accuracy = accuracy_score(
    y_test,
    ada_prediction
)

print("\n=====")
print("ADABOOST RESULT")
print("=====")

print(
    "Accuracy:",
    round(ada_accuracy * 100, 2),
    "%"
)

improvement = (
    ada_accuracy - weak_accuracy
) * 100

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```

```
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print("\n=====")
print("MODEL COMPARISON")
print("=====")

print(
    "Weak Classifier Accuracy:",
    round(weak_accuracy * 100, 2),
    "%"
)

print(
    "AdaBoost Accuracy:",
    round(ada_accuracy * 100, 2),
    "%"
)

print(
    "Improvement:",
    round(improvement, 2),
    "percentage points"
)

print("\n=====")
print("CLASSIFICATION REPORT")
print("=====")

print(
    classification_report(

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```

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File Edit Format Run Options Window Help

y_test,
ada_prediction,
target_names=[
    "Legitimate",
    "Phishing"
],
zero_division=0
)
)

cm = confusion_matrix(
    y_test,
    ada_prediction
)

print("\nConfusion Matrix:")
print(cm)

plt.figure(figsize=(7, 6))

plt.imshow(cm)

plt.title(
    "AdaBoost Confusion Matrix"
)

plt.xlabel(
    "Predicted Class"
)

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plt.ylabel(
    "Actual Class"
)

plt.xticks(
    [0, 1],
    ["Legitimate", "Phishing"]
)

plt.yticks(
    [0, 1],
    ["Legitimate", "Phishing"]
)

for i in range(2):
    for j in range(2):
        plt.text(
            j,
            i,
            str(cm[i, j]),
            ha="center",
            va="center"
        )

plt.colorbar()

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```

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plt.tight_layout()
plt.show()

models = [
    "Weak Classifier",
    "AdaBoost"
]

accuracies = [
    weak_accuracy * 100,
    ada_accuracy * 100
]

plt.figure(figsize=(8, 5))

plt.bar(
    models,
    accuracies
)

plt.title(
    "Weak Classifier vs AdaBoost"
)

plt.xlabel(
    "Model"
)
```

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```
T092 Practical no 6.py - C:/Users/Vinod Mali/Desktop/AI/T092 Practical no 6.py (3.14.8)
File Edit Format Run Options Window Help

plt.ylabel(
    "Accuracy (%)"
)

for i, value in enumerate(accuracies):

    plt.text(
        i,
        value,
        f"{value:.2f}%",
        ha="center",
        va="bottom"
    )

plt.ylim(0, 100)
plt.tight_layout()
plt.show()

print("\n=====")
print("ADABOOST PRACTICAL COMPLETED")
print("=====")

print(
    "Weak Classifier:",
    round(weak_accuracy * 100, 2),
)
```

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```
round(weak_accuracy * 100, 2),
"%")
)
print(
    "AdaBoost:",
    round(ada_accuracy * 100, 2),
    "%")
)
print(
    "Improvement:",
    round(improvement, 2),
    "percentage points"
)
print("=====")
```

Output :-

```
"IDLE Shell 3.14.6"
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Vinod Mali/Desktop/Ai/T092 Practical no 6.py =====
Dataset loaded successfully!

Dataset Shape:
(10000, 50)

First 5 Rows:
  id  NumDots  ...  PctExtNullSelfRedirectHyperlinksRT  CLASS_LABEL
0  1    3  ...                1                1
1  2    3  ...                1                1
2  3    3  ...                0                1
3  4    3  ...               -1                1
4  5    3  ...               -1                1

[5 rows x 50 columns]

Shape after cleaning:
(10000, 50)

Target Column:
CLASS_LABEL

Number of Features:
49

Target Classes:
CLASS_LABEL
```

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```
"IDLE Shell 3.14.0"
File Edit Shell Debug Options Window Help
1 5000
0 5000
Name: count, dtype: int64

Training Samples:
8000

Testing Samples:
2000

=====
INDIVIDUAL WEAK CLASSIFIER
=====
Accuracy: 100.0 %

Training AdaBoost...
AdaBoost trained successfully!

=====
ADABOOST RESULT
=====
Accuracy: 100.0 %

=====
MODEL COMPARISON
=====
Weak Classifier Accuracy: 100.0 %
AdaBoost Accuracy: 100.0 %
Improvement: 0.0 percentage points

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```

```
=====
CLASSIFICATION REPORT
=====
      precision    recall  f1-score   support

 Legitimate      1.00      1.00      1.00     1000
  Phishing      1.00      1.00      1.00     1000

 accuracy              1.00     2000
 macro avg      1.00      1.00      1.00     2000
 weighted avg    1.00      1.00      1.00     2000

Confusion Matrix:
[[1000  0]
 [  0 1000]]
S

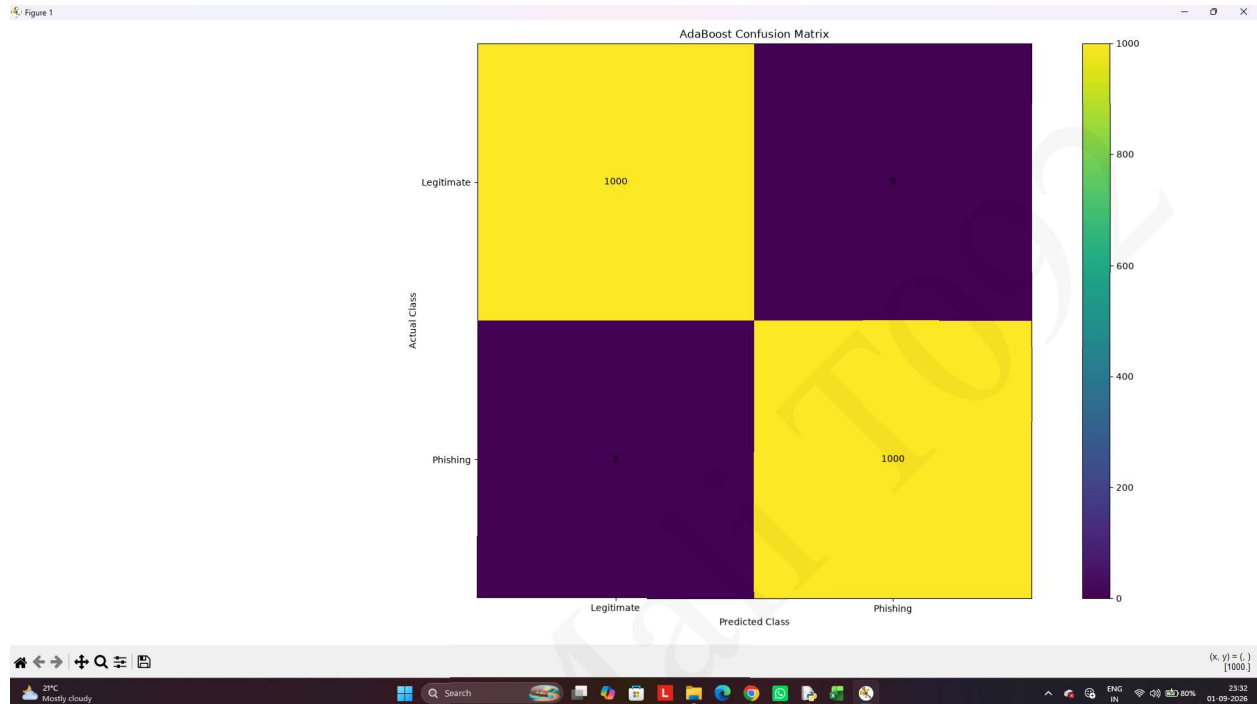
Ln: 77 Col: 1
```

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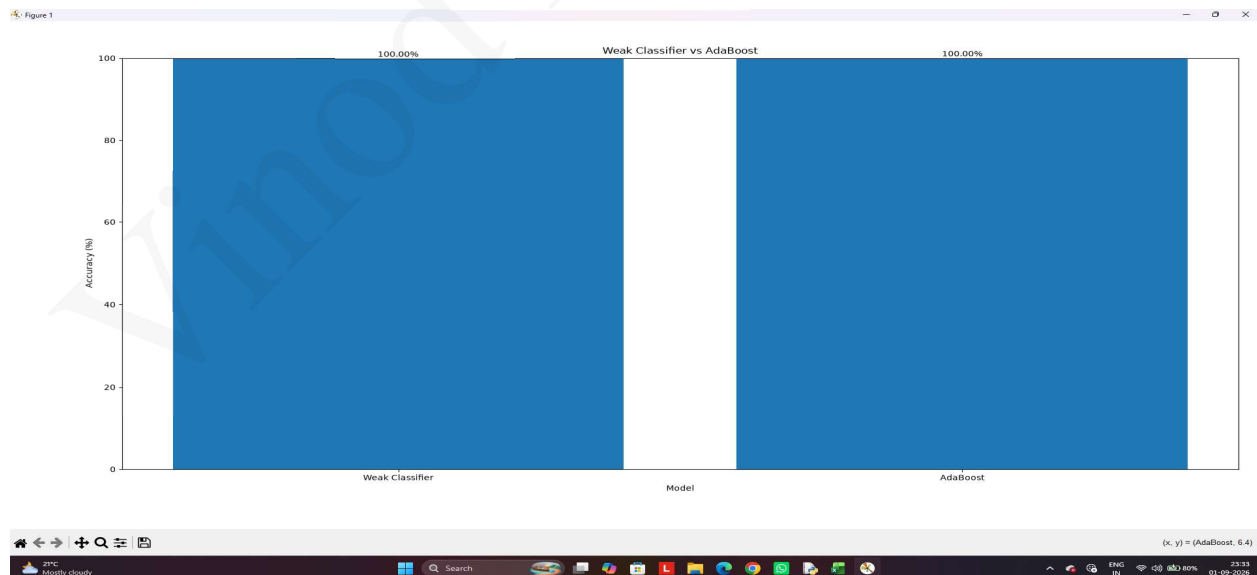
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Graphs:-

AdaBoost Confusion Matrix :-



Weak classifier Vs AdaBoost :-



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Observations :-

1. Confusion Matrix:

- **The diagonal values represent correct predictions of legitimate and phishing websites.**
- **The off-diagonal values represent incorrect predictions.**
- **The model shows good classification performance if the diagonal values are higher.**

2. Accuracy Comparison Graph:

- **The AdaBoost model generally has higher accuracy than the individual weak classifier.**
- **This shows that combining multiple weak classifiers improves the overall performance.**
- **Therefore, AdaBoost is more effective than a single Decision Tree (weak classifier).**

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